

Raritone Project Backend — Daily Task Documentation

Auth-security (security improvements)

This backend project is built using **Node.js, Express, and MongoDB** to support an e-commerce system with authentication, cart, orders, and measurement features.

Features Implemented Today

1. Authentication System

- User registration API
- User login API
- JWT-based authentication setup (structure ready)
- Protected route middleware (`protect`)
- Role-based access middleware (`authorizeRoles`)

2. Cart Module

- Add items to cart
- Get user cart
- Remove items from cart
- Protected routes (authentication required)

3. Order Module

- Create order
- Get all orders
- Get single order by ID
- Update order status
- Delete order
- User-based protected access

4. Measurement Module

- Save user measurements (create/update in one API)
- Get user measurement
- Delete measurement
- Linked to authenticated user (`req.user.id`)

5. Middleware System

- JWT authentication middleware (`protect`)
- Role-based authorization middleware (`authorizeRoles`)
- Error handling structure added
- Route protection implemented across modules

6. Server Setup

- Express server setup
- MongoDB connection using Mongoose
- Helmet for security
- Rate limiting implemented
- Centralized route handling

7. API Routes Structure

```
/api/auth  
/api/cart  
/api/order  
/api/measurement
```

8. Project Structure

```
controllers/  
routes/  
middlewares/  
models/  
server.js  
.env
```

Tech Stack

- Node.js
- Express.js
- MongoDB (Mongoose)
- JWT (Authentication)
- Helmet (Security)
- Express Rate Limit

Issues Faced & Fixed

✖ Module not found errors

✓ Fixed incorrect folder paths (middleware → middlewares)

✖ Handler must be a function error

✓ Fixed wrong controller exports/import mismatch

✖ authMiddleware undefined issue

✓ Replaced with correct protect middleware

✖ Route integration issues

✓ Standardized all routes to use protect

GitHub Update

- Created branch: auth-security
- Pushed backend code to GitHub repository:

👉 <https://github.com/Nandini-Bhimineni/Raritone-Project-Backend>

Current Status

- ✓ Backend structure complete
- ✓ Core APIs implemented
- ✓ Middleware system working

✓ MongoDB connected

✓ GitHub branch updated

Done by

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